

1. NAME OF THE MEDICINAL PRODUCT

Cervarix suspension for injection in pre-filled syringe
Human Papillomavirus vaccine [Types 16, 18] (Recombinant, adjuvanted, adsorbed)

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

1 dose (0.5 ml) contains:

Human Papillomavirus ¹ type 16 L1 protein ^{2,3,4}	20 micrograms
Human Papillomavirus ¹ type 18 L1 protein ^{2,3,4}	20 micrograms

¹Human Papillomavirus = HPV

² adjuvanted by AS04 containing: 3-O-desacyl-4'-monophosphoryl lipid A (MPL) ³	50 micrograms
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³ adsorbed on aluminium hydroxide, hydrated (Al(OH) ₃)	0.5 milligrams Al ³⁺ in total
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⁴L1 protein in the form of non-infectious virus-like particles (VLPs) produced by recombinant DNA technology using a Baculovirus expression system which uses Hi-5 Rix4446 cells derived from *Trichoplusia ni*.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Suspension for injection.

4. CLINICAL INFORMATION**4.1 Therapeutic indications**

CERVARIX is indicated from 10 to 45 years of age for the prevention of premalignant ano-genital lesions (cervical, vulvar, vaginal and anal) and cervical and anal cancers caused by oncogenic Human Papillomaviruses (HPV) type 16 and 18.

4.2 Posology and method of administrationPosology

The vaccination schedule depends on the age of the subject.

Age at the time of first injection	Immunization and schedule
10 to and including 14 years*	Two doses each of 0.5 ml. The second dose given between 5 and 13 months after the first dose
From 15 years and above	Three doses each of 0.5 ml at 0, 1, 6 months**

*If the second vaccine dose is administered before the 5th month after the first dose, a third dose should always be administered.

**If flexibility in the vaccination schedule is necessary, the second dose can be administered between 1 month and 2.5 months after the first dose and the third dose between 5 and 12 months after the first dose.

The need for a booster dose has not been established (see section 5.1).

It is recommended that subjects who receive a first dose of Cervarix complete the vaccination course with Cervarix (see section 4.4).

Paediatric population

Cervarix is not recommended for use in children below 10 years of age due to lack of data on safety and immunogenicity in this age-group.

Method of administration

Cervarix is for intramuscular injection in the deltoid region (see also sections 4.4 and 4.5).

4.3 Contraindications

Hypersensitivity to the active substances or to any of the excipients listed in section 6.1.

4.4 Special warnings and precautions for use

As with all injectable vaccines, appropriate medical treatment and supervision should always be readily available in case of a rare anaphylactic reaction following the administration of the vaccine.

Syncope (fainting) can occur following, or even before, any vaccination especially in adolescents as a psychogenic response to the needle injection. This can be accompanied by several neurological signs such as transient visual disturbance, paraesthesia and tonic-clonic limb movements during recovery. It is important that procedures are in place to avoid injury from faints.

Administration of Cervarix should be postponed in subjects suffering from an acute severe febrile illness. However, the presence of a minor infection, such as a cold, is not a contraindication for immunisation.

Cervarix should under no circumstances be administered intravascularly or intradermally. No data are available on subcutaneous administration of Cervarix.

As with other vaccines administered intramuscularly, Cervarix should be given with caution to individuals with thrombocytopenia or any coagulation disorder since bleeding may occur following an intramuscular administration to these subjects.

As with any vaccine, a protective immune response may not be elicited in all vaccines.

Cervarix will only protect against diseases that are caused by HPV types 16 and 18 and to some extent against diseases caused by certain other oncogenic related HPV types (see section 5.1). Therefore, appropriate precautions against sexually transmitted diseases should continue to be used.

Cervarix is for prophylactic use only and has no effect on active HPV infections or established clinical disease. Cervarix has not been shown to have a therapeutic effect. The vaccine is therefore not indicated for treatment of cervical cancer or cervical intraepithelial neoplasia (CIN). It is also not intended to prevent progression of other established HPV-related lesions or existing HPV infections with vaccine

or non-vaccine types (see section 5.1 “Efficacy against HPV-16/18 in women with evidence of HPV-16 or HPV-18 infection at study entry.”).

Vaccination is not a substitute for routine cervical screening. Since no vaccine is 100% effective and Cervarix will not provide protection against every HPV type, or against existing HPV infections, routine cervical screening remains critically important and should follow local recommendations.

Duration of protection has not fully been established. Timing and need of booster dose(s) has not been established.

Except for asymptomatic human immunodeficiency virus (HIV) infected subjects for whom immunogenicity data are available (see section 5.1), there are no data on the use of Cervarix in subjects with impaired immune responsiveness such as patients receiving immunosuppressive treatment. As with other vaccines, an adequate immune response may not be elicited in these individuals.

There are no safety, immunogenicity or efficacy data to support interchangeability of Cervarix with other HPV vaccines.

4.5 Interaction with other medicinal products and other forms of interaction

In all clinical trials individuals who had received immunoglobulin or blood products within 3 months prior to the first vaccine dose were excluded.

Use with other vaccines

Cervarix may be administered concomitantly with a combined booster vaccine containing diphtheria (d), tetanus (T) and pertussis [acellular] (pa) with or without inactivated poliomyelitis (IPV), (dTpa, dTpa-IPV vaccines), with no clinically relevant interference with antibody response to any of the components of either vaccine. The sequential administration of combined dTpa-IPV followed by Cervarix one month later tended to elicit lower anti-HPV-16 and anti-HPV-18 GMTs as compared to Cervarix alone. The clinical relevance of this observation is not known.

Cervarix may also be administered concomitantly with meningococcal serogroups A, C, W-135, Y tetanus toxoid conjugate vaccine (MenACWY-TT); with combined hepatitis A (inactivated) and hepatitis B (rDNA) vaccine (Twinrix) or with hepatitis B (rDNA) vaccine (Engerix B).

Administration of Cervarix at the same time as Twinrix has shown no clinically relevant interference in the antibody response to the HPV and hepatitis A antigens. Anti-HBs geometric mean antibody concentrations were significantly lower on co-administration, but the clinical relevance of this observation is not known since the seroprotection rates remain unaffected. The proportion of subjects reaching anti-HBs ≥ 10 mIU/ml was 98.3% for concomitant vaccination and 100% for Twinrix given alone. Similar results were observed when Cervarix was given concomitantly with Engerix B with 97.9% of subjects reaching anti-HBs ≥ 10 mIU/ml compared to 100% for Engerix B given alone.

If Cervarix is to be given at the same time as another injectable vaccine, the vaccines should always be administered at different injection sites.

Use with hormonal contraceptive

In clinical efficacy studies, approximately 60% of women who received Cervarix used hormonal contraceptives. There is no evidence that the use of hormonal contraceptives has an impact on the efficacy of Cervarix.

Use with systemic immunosuppressive medicinal products

As with other vaccines it may be expected that, in patients receiving immunosuppressive treatment, an adequate response may not be elicited.

4.6 Fertility, pregnancy and lactation

Pregnancy

Specific studies of the vaccine in pregnant women were not conducted. Data in pregnant women collected as part of pregnancy registries, epidemiological studies and inadvertent exposure during clinical trials are insufficient to conclude whether or not vaccination with Cervarix affects the risk of adverse pregnancy outcomes including spontaneous abortion.

However, during the clinical development program, a total of 10,476 pregnancies were reported including 5,387 in women who had received Cervarix. Overall, the proportions of pregnant subjects who experienced specific outcomes (e.g., normal infant, abnormal infants including congenital anomalies, premature birth, and spontaneous abortion) were similar between treatment groups.

Animal studies do not indicate direct or indirect harmful effects with respect to fertility, pregnancy, embryonal/foetal development, parturition or post-natal development (see section 5.3).

As a precautionary measure, it is preferable to avoid the use of Cervarix during pregnancy. Women who are pregnant or trying to become pregnant, are advised to postpone or interrupt vaccination until completion of pregnancy.

Breast-feeding

The effect on breast-fed infants of the administration of Cervarix to their mothers has not been evaluated in clinical studies.

Cervarix should only be used during breast-feeding when the possible advantages outweigh the possible risks.

Fertility

No fertility data are available.

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive or use machines have been performed.

4.8 Undesirable effects

Summary of safety profile

In clinical studies that enrolled girls and women aged from 10 up to 72 years (of which 79.2% were aged 10-25 years at the time of enrolment), Cervarix was administered to 16,142 subjects whilst 13,811 subjects received control. These subjects were followed for serious adverse events over the entire study period. In a pre-defined subset of subjects (Cervarix = 8,130 versus control = 5,786), adverse events were followed for 30 days after each injection. In two clinical studies that enrolled males aged 10 to 18 years, 2,617 males received Cervarix and were followed-up with active safety surveillance.

The most common adverse reaction observed after vaccine administration was injection site pain which occurred after 78% of all doses. The majority of these reactions were of mild to moderate severity and were not long lasting.

List of adverse reactions

Adverse reactions considered as being at least possibly related to vaccination have been categorised by frequency.

Frequencies are reported as:

Very common ($\geq 1/10$)

Common ($\geq 1/100$ to $< 1/10$)

Uncommon ($\geq 1/1,000$ to $< 1/100$)

- *Clinical trial data*

Infections and infestations:

Uncommon: upper respiratory tract infection

Nervous system disorders:

Very common: headache

Uncommon: dizziness

Gastrointestinal disorders:

Common: gastrointestinal symptoms including nausea, vomiting, diarrhoea and abdominal pain

Skin and subcutaneous tissue disorders:

Common: itching/pruritus, rash, urticaria

Musculoskeletal and connective tissue disorders:

Very common: myalgia

Common: arthralgia

General disorders and administration site conditions:

Very common: injection site reactions including pain, redness, swelling; fatigue

Common: fever ($\geq 38^{\circ}\text{C}$)

Uncommon: other injection site reactions such as induration, local paraesthesia

A similar safety profile has been observed in subjects with prior or current HPV infection as compared to subjects negative for oncogenic HPV DNA or seronegative for HPV-16 and HPV-18 antibodies.

- *Post marketing data*

Because these events were reported spontaneously, it is not possible to reliably estimate their frequency.

Blood and lymphatic system disorders

Lymphadenopathy

Immune system disorders

Allergic reactions (including anaphylactic and anaphylactoid reactions), angioedema

Nervous system disorders

Syncope or vasovagal responses to injection, sometimes accompanied by tonic-clonic movements (see section 4.4)

4.9 Overdose

No case of overdose has been reported.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmaco-therapeutic group: Vaccines, Papillomavirus vaccines, ATC code: J07BM02

Mechanism of action

Persistent infection with oncogenic HPV types has been demonstrated to be responsible for virtually all cases of cervical cancer worldwide.

Cervarix is a non-infectious recombinant vaccine prepared from the highly purified virus-like particles (VLPs) of the major capsid L1 protein of oncogenic HPV types 16 and 18. Since the VLPs contain no viral DNA, they cannot infect cells, reproduce or cause disease. Animal studies have shown that the efficacy of L1 VLP vaccines is largely mediated by the development of an humoral immune response and cell-mediated immune memory.

Cervarix is adjuvanted with AS04 which has been shown in clinical trials to induce a higher and long lasting immune response compared to the same antigens adjuvanted with aluminium salt $[Al(OH)_3]$ alone.

Invasive cervical cancer includes squamous cervical carcinoma (84%) and adenocarcinoma (16%, up to 20% in developed countries with screening programs).

HPV-16 and HPV-18 are responsible for approximately 70% of cervical cancers, 80% of vulvar and vaginal cancers, 90% of anal cancers, 70% of HPV related high-grade vulvar (VIN 2/3) and vaginal intraepithelial neoplasia (VaIN 2/3) and 78% of HPV related high-grade anal (AIN 2/3) intraepithelial neoplasia across all regions worldwide.

Other oncogenic HPV types (HPV-31, -33, -35, -39, -45, -51, -52, -56, -58, -59, -66, -68) can also cause ano-genital cancers. HPV-16, -18, -45 and -31 are the 4 most common types identified in squamous cervical carcinoma (approximately 76%) and adenocarcinoma (approximately 91%).

Evidence of Anamnestic (Immune Memory) Response

The administration of a challenge dose after a mean of 6.8 years following the first vaccination elicited an anamnestic immune response to HPV-16 and HPV-18 (by ELISA and pseudovirion-based neutralizing assay) at day 7. One month after the challenge dose, GMTs exceeded those observed one month after the primary vaccination course.

An anamnestic response was also observed for the related types HPV-31 and HPV-45 by ELISA.

Clinical studies

Clinical efficacy in women aged 15 to 25 years

The efficacy of Cervarix was assessed in two controlled, double-blind, randomised Phase II and III clinical trials that included a total of 19,778 women aged 15 to 25 years.

The phase II trial (study 001/007) enrolled only women who:

- Were tested negative for oncogenic HPV DNA of types 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, 66 and 68
- Were seronegative for HPV-16 and HPV-18 and
- Had normal cytology

The primary efficacy endpoint was incident infection with HPV-16 and/or HPV-18. Twelve-month persistent infection was evaluated as additional efficacy endpoint.

The phase III trial (study 008) enrolled women without pre-screening for the presence of HPV infection, i.e. regardless of baseline cytology and HPV serological and DNA status.

The primary efficacy endpoint was CIN2+ associated with HPV-16 and/or HPV-18 (HPV-16/18). Cervical Intraepithelial Neoplasia (CIN) grade 2 and 3 (CIN2/3) and cervical adenocarcinoma in situ (AIS) were used in the clinical trials as surrogate markers for cervical cancer. The secondary endpoints included 6- and 12-month persistent infection.

Persistent infection that lasts for at least 6 months has also been shown to be a relevant surrogate marker for cervical cancer.

Prophylactic efficacy against HPV-16/18 infection in a population naïve to oncogenic HPV types

Women (N=1,113) were vaccinated in study 001 and evaluated for efficacy up to month 27. A subset of women (N=776) vaccinated in study 001 was followed in study 007 up to 6.4 years (approximately 77 months) after the first dose (mean follow-up of 5.9 years). There were five cases of 12-month persistent HPV-16/18 infection (4 HPV-16; 1 HPV-18) in the control group and one HPV-16 case in the vaccine group in study 001. In study 007 the efficacy of Cervarix against 12-month persistent HPV-16/18 infection was 100% (95% CI: 80.5; 100). There were sixteen cases of persistent HPV-16 infection, and five cases of persistent HPV-18 infection, all in the control group.

In study HPV-023, subjects from the Brazilian cohort (N=437) of study 001/007 were followed up to a mean of 8.9 years (standard deviation 0.4 years) after the first dose. At study completion, there were no cases of infection or histopathological lesions associated with HPV-16 or HPV-18 in the vaccine group in study HPV-023. In the placebo group, there were 4 cases of 6-month persistent infection and 1 case of 12-month persistent infection. The study was not powered to demonstrate a difference between the vaccine and the placebo group for these endpoints.

Prophylactic efficacy against HPV-16/18 in women naïve to HPV-16 and/or HPV-18

In study HPV-008, the primary analyses of efficacy were performed on the According to Protocol cohort (ATP cohort: including women who received 3 vaccine doses and were DNA negative and seronegative at month 0 and DNA negative at month 6 for the HPV type considered in the analysis). This cohort included women with normal or low-grade cytology at baseline and excluded only women with high-grade cytology (0.5% of the total population). Case counting for the ATP cohort started on day 1 after the third dose of vaccine.

Overall, 74% of women enrolled were naïve to both HPV-16 and HPV-18 (i.e. DNA negative and seronegative at study entry).

Two analyses of study HPV-008 have been performed: an event-triggered analysis performed once at least 36 CIN2+ cases associated with HPV-16/18 were accrued in the ATP cohort and an end-of study analysis.

Vaccine efficacy against the primary endpoint CIN2+ is presented in Table 1. In a supplemental analysis, the efficacy of Cervarix was evaluated against HPV-16/18-related CIN3+.

Table 1: Vaccine efficacy against high grade cervical lesions associated with HPV-16/18 (ATP cohort)

HPV-16/18 endpoint	ATP cohort ⁽¹⁾		
	End of study analysis ⁽³⁾		
	Cervarix (N = 7338)	Control (N = 7305)	% Efficacy (95% CI)
	n ⁽²⁾	n	
CIN2+	5	97	94.9% (87.7;98.4)
CIN3+	2	24	91.7% (66.6;99.1)
N = number of subjects included in each group n = number of cases			

⁽¹⁾ ATP: includes women who received 3 doses of vaccine, were DNA negative and seronegative at month 0 and DNA negative at month 6 to the relevant HPV type (HPV-16 or HPV-18)

⁽²⁾ including 4 cases of CIN2+ and 2 cases of CIN3+ in which another oncogenic HPV type was identified in the lesion, concomitantly with HPV-16 or HPV-18. These cases are excluded in the HPV type assignment analysis (see under Table).

⁽³⁾ mean follow-up of 40 months post dose 3

At the event-triggered analysis the efficacy was 92.9% (96.1% CI:79.9;98.3) against CIN2+ and 80% (96.1% CI: 0.3;98.1) against CIN3+. In addition, statistically significant vaccine efficacy against CIN2+ associated with HPV-16 and HPV-18 individually was demonstrated.

Further investigation of the cases with multiple HPV types considered the HPV types detected by Polymerase Chain Reaction (PCR) in at least one of the two preceding cytology samples, in addition to types detected in the lesion to distinguish the HPV type(s) most likely responsible to the lesion (HPV type assignment). This post-hoc analysis excluded cases (in the vaccine group and in the control group) which were not considered to be causally associated with HPV-16 or HPV-18 infections acquired during the trial.

Based on the HPV type assignment post-hoc analysis, there was 1 CIN2+ case in the vaccine group versus 92 cases in the control group (Efficacy 98.9% (95% CI: 93.8;100)) and no CIN3+ case in the vaccine group versus 22 cases in the control group (Efficacy 100% (95% CI: 81.8;100)) at the end of study analysis.

In the event-triggered analysis, vaccine efficacy against CIN1 associated with HPV 16/18 observed in the ATP cohort was 94.1% (96.1% CI: 83.4;98.5). Vaccine efficacy against CIN1+ associated with HPV 16/18 observed in the ATP cohort was 91.7% (96.1% CI: 82.4;96.7). At the end of study analysis, vaccine efficacy against CIN1 associated with HPV 16/18 observed in the ATP cohort was 92.8% (95% CI: 87.1;96.4).

At end of study analysis, there were 2 cases of VIN2+ or VaIN2+ in the vaccine group and 7 cases in the control group in the ATP cohort associated with HPV-16 or HPV-18. The study was not powered to demonstrate a difference between the vaccine and the control group for these endpoints.

Vaccine efficacy against virological endpoints (6-month and 12-month persistent infection) associated with HPV-16/18 observed in the ATP cohort at the end of study is presented in Table 2.

Table 2: Vaccine efficacy against virological endpoints associated with HPV-16/18 (ATP cohort)

HPV-16/18 endpoint	ATP cohort ⁽¹⁾		
	End of study analysis ⁽²⁾		
	Cervarix (N = 7338)	Control (N = 7305)	% Efficacy (95% CI)
	n/N	n/N	
6-month persistent infection	35/7182	588/7137	94.3% (92.0;96.1)
12-month persistent infection	26/7082	354/7038	92.9% (89.4;95.4)
N = number of subjects included in each group n = number of cases ⁽¹⁾ ATP: includes women who received 3 doses of vaccine, were DNA negative and seronegative at month 0 and DNA negative at month 6 to the relevant HPV type (HPV-16 or HPV-18) ⁽²⁾ mean follow-up of 40 months post dose 3			

The efficacy results at the event-triggered analysis were 94.3% (96.1% CI:91.5;96.3) against 6-month persistent infection and 91.4% (96.1% CI: 89.4;95.4) against 12-month persistent infection.

Efficacy against HPV-16/18 in women with evidence of HPV-16 or HPV-18 infection at study entry.

There was no evidence of protection from disease caused by the HPV types for which subjects were HPV DNA positive at study entry. However, individuals already infected (HPV DNA positive) with one of the vaccine-related HPV types prior to vaccination were protected from clinical disease caused by the other vaccine HPV type.

Efficacy against HPV types 16 and 18 in women with and without prior infection or disease.

The Total Vaccinated Cohort (TVC) included all subjects who received at least one dose of the vaccine, irrespective of their HPV DNA status, cytology and serostatus at baseline. This cohort included women with or without current and/or prior HPV infection. Case counting for the TVC started on day 1 after the first dose.

The efficacy estimates are lower in the TVC as this cohort includes women with pre-existing infections/lesions, which are not expected to be impacted by Cervarix.

The TVC may approximate to the general population of women in the age range of 15-25 years.

Vaccine efficacy against high grade cervical lesions associated with HPV-16/18 observed in TVC at end of study is presented in Table 3.

Table 3: Vaccine efficacy against high grade cervical lesions associated with HPV-16/18 (TVC)

HPV-16/18 endpoint	TVC ⁽¹⁾		
	End of study analysis ⁽²⁾		
	Cervarix (N = 8694)	Control (N = 8708)	% Efficacy (95% CI)
	n	n	
CIN2+	90	228	60.7% (49.6;69.5)
CIN3+	51	94	45.7% (22.9;62.2)
N = number of subjects included in each group n = number of cases ⁽¹⁾ TVC: includes all vaccinated subjects (who received at least one dose of vaccine) irrespective of HPV DNA status, cytology and serostatus at baseline. This cohort includes women with pre-existing infections/lesions ⁽²⁾ mean follow-up of 44 months post dose 1			

Vaccine efficacy against virological endpoints (6-month and 12-month persistent infection) associated with HPV-16/18 observed in TVC at end of study is presented in Table 4.

Table 4: Vaccine efficacy against virological endpoints associated with HPV-16/18 (TVC)

HPV-16/18 endpoint	TVC ⁽¹⁾		
	End of study analysis ⁽²⁾		
	Cervarix	Control	% Efficacy (95% CI)
	n/N	n/N	
6-month persistent infection	504/8863	1227/8870	60.9% (56.6;64.8)
12-month persistent infection	335/8648	767/8671	57.5% (51.7;62.8)
N = number of subjects included in each group n = number of cases ⁽¹⁾ TVC: includes all vaccinated subjects (who received at least one dose of vaccine) irrespective of HPV DNA status, cytology and serostatus at baseline. ⁽²⁾ mean follow-up of 44 months post dose 1			

Overall impact of the vaccine on cervical HPV disease burden

In study HPV-008, the incidence of high grade cervical lesions was compared between the placebo and vaccine group irrespective of the HPV DNA type in the lesion. In the TVC and TVC-naïve cohorts, the vaccine's efficacy was demonstrated against high-grade cervical lesions at end of study (Table 5). The TVC-naïve is a subset of the TVC that includes women with normal cytology, and who were HPV DNA negative for 14 oncogenic HPV types and seronegative for HPV-16 and HPV-18 at baseline.

Table 5: Vaccine efficacy against high-grade cervical lesions irrespective of the HPV DNA type in the lesion

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	End of study analysis ⁽³⁾				
	Cervarix		Control		% Efficacy (95% CI)
	N	Cases	N	Cases	
CIN2+					
TVC-naïve ⁽¹⁾	5466	61	5452	172	64.9% (52.7;74.2)
TVC ⁽²⁾	8694	287	8708	428	33.1% (22.2;42.6)
CIN3+					
TVC-naïve ⁽¹⁾	5466	3	5452	44	93.2% (78.9;98.7)
TVC ⁽²⁾	8694	86	8708	158	45.6% (28.8;58.7)
N = number of subjects included in each group					
⁽¹⁾ TVC naïve: includes all vaccinated subjects (who received at least one dose of vaccine) who had normal cytology, were HPV DNA negative for 14 oncogenic HPV types and seronegative for HPV-16 and HPV-18 at baseline.					
⁽²⁾ TVC: includes all vaccinated subjects (who received at least one dose of vaccine) irrespective of HPV DNA status, cytology and serostatus at baseline.					
⁽³⁾ mean follow-up of 44 months post dose 1					

At the end of study analysis, Cervarix reduced definitive cervical therapy procedures (includes loop electrosurgical excision procedure [LEEP], cold-knife Cone, and laser procedures) by 70.2% (95% CI: 57.8;79.3) in TVC-naïve and 33.2% (95% CI: 20.8;43.7) in TVC.

Cross-protective efficacy

The cross-protective efficacy of Cervarix against histopathological and virological endpoints (persistent infection) has been evaluated in study HPV-008 for 12 non-vaccine oncogenic HPV types. The study was not powered to assess efficacy against disease caused by individual HPV types. The analysis against the primary endpoint was confounded by multiple co-infections in the CIN2+ lesions. Unlike histopathological endpoints, virological endpoints are less confounded by multiple infections.

HPV-31, 33 and 45 showed consistent cross-protection for 6-month persistent infection and CIN2+ endpoints in all study cohorts.

End of study vaccine efficacy against 6-month persistent infection and CIN2+ associated with individual non-vaccine oncogenic HPV types is presented in Table 6 (ATP cohort).

Table 6: Vaccine efficacy for non-vaccine oncogenic HPV types

ATP ⁽¹⁾						
HPV type	6-month persistent infection			CIN2+		
	Cervarix	Control	% Efficacy (95% CI)	Cervarix	Control	% Efficacy (95% CI)
	n	n		n	n	
HPV-16 related types (A9 species)						
HPV-31	58	247	76.8% (69.0;82.9)	5	40	87.5% (68.3;96.1)
HPV-33	65	117	44.8% (24.6;59.9)	13	41	68.3% (39.7;84.4)
HPV-35	67	56	-19.8% (<0;17.2)	3	8	62.5% (<0;93.6)
HPV-52	346	374	8.3% (<0;21.0)	24	33	27.6% (<0;59.1)
HPV-58	144	122	-18.3% (<0;7.7)	15	21	28.5% (<0;65.7)
HPV-18 related types (A7 species)						
HPV-39	175	184	4.8% (<0;23.1)	4	16	74.9% (22.3;93.9)
HPV-45	24	90	73.6% (58.1;83.9)	2	11	81.9% (17.0;98.1)
HPV-59	73	68	-7.5% (<0;23.8)	1	5	80.0% (<0;99.6)
HPV-68	165	169	2.6% (<0;21.9)	11	15	26.8% (<0;69.6)
Other types						
HPV-51	349	416	16.6% (3.6;27.9)	21	46	54.4% (22.0;74.2)
HPV-56	226	215	-5.3% (<0;13.1)	7	13	46.1% (<0;81.8)
HPV-66	211	215	2.3% (<0;19.6)	7	16	56.4% (<0;84.8)

n= number of cases

⁽¹⁾ ATP: includes women who received 3 doses of vaccine, were DNA negative at month 0 and at month 6 to the relevant HPV type.

The limits of the confidence interval around the vaccine efficacy were calculated. When the value zero is included, i.e. when the lower limit of the CI is <0, the efficacy is not considered statistically significant.

The efficacy against CIN3 was only demonstrated for HPV-31 and there was no evidence of protection against AIS for any of the HPV types.

Clinical efficacy in women aged 26 years and older

The efficacy of Cervarix was assessed in a double-blind, randomised Phase III clinical trial (HPV-015) that included a total of 5,778 women aged 26-72 years (median: 37.0 years). The study was conducted in North America, Latin America, Asia Pacific and Europe. Final analysis was performed at study conclusion, 7 years after 1st vaccination.

The primary endpoint was a combination of a virological and a histopathological endpoint: HPV-16/18 related 6-month persistent infection and/or CIN1+. The primary analyses of efficacy were performed on the ATP cohort for efficacy and the TVC which included a subset of up to 15% of women with a history of HPV-associated infection or disease (defined as two or more abnormal smears in sequence, abnormal colposcopy, or biopsy or treatment of the cervix after abnormal smear or colposcopy findings). Inclusion of this subset allowed assessment of prophylactic efficacy in a population that is

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thought to reflect a real-world setting, as adult women are the age group generally targeted for cervical screening.

Vaccine efficacy at study conclusion is summarised in the following table.

There is no evidence whether prevention of persistent infection that lasts for at least 6 months is a relevant surrogate marker for cervical cancer prevention in women aged 26 years and above.

Table 7 - Vaccine efficacy at study conclusion in study HPV-015

Endpoint	ATP ⁽¹⁾			TVC ⁽²⁾		
	Cervari x	Control	% Efficacy (96.2% CI)	Cervarix	Control	% Efficacy (96.2% CI)
	n/N	n/N		n/N	n/N	
HPV-16/18						
6M PI and/or CIN1+	7/1,852	71/1,818	90.5% (78.6; 96.5)	93/2,768	209/2,778	56.8% (43.8; 67.0)
6M PI	6/1,815	67/1,786	91.4% (79.4; 97.1)	74/2,762	180/2,775	60% (46.4; 70.4)
CIN2+	1/1,852	6/1,818	83.7% (<0.0; 99.7)	33/2,733	51/2,735	35.8% (<0.0; 61.0)
ASC-US+	3/1,852	47/1,818	93.8% (79.9; 98.9)	38/2,727	114/2,732	67.3% (51.4; 78.5)
6M PI in subjects seropositive at baseline only	3/851	13/837	78% (15.0; 96.4)	42/1,211	65/1,192	38.7% (6.3; 60.4)
Cross protective efficacy						
HPV-31 6M PI	10/2,073	29/2,090	65.8% (24.9; 85.8)	51/2,762	71/2,775	29% (<0.0; 52.5)
HPV-45 6M PI	9/2,106	30/2,088	70.7% (34.2; 88.4)	22/2,762	60/2,775	63.9% (38.6; 79.6)
HPV-31 ASC-US+	5/2,117	23/2,127	78.4% (39.1; 94.1)	34/2,727	55/2,732	38.7% (2.0; 62.3)
HPV-45 ASC-US+	5/2,150	23/2,125	78.7% (40.1; 94.1)	13/2,727	38/2,732	66.1% (32.7; 84.1)
N = number of subjects in each group n = number of subjects reporting at least one event in each group 6M PI = 6-month persistent infection CI = Confidence Interval ASC-US= Atypical Cells of Undetermined Significance (abnormal cytology) ⁽¹⁾ 3 doses of vaccine, DNA negative and seronegative at month 0 (unless specified) and DNA negative at month 6 for the relevant HPV type (HPV-16 and/or HPV-18) ⁽²⁾ at least one dose of vaccine, irrespective of HPV DNA and serostatus (unless specified) at month 0. Includes 15% of subjects with previous history of HPV disease/infection						

Efficacy against ≥ASC-US (abnormal cytology) associated with oncogenic non-vaccine types was 37.2% (96.2% CI [21.3; 50.1]) (ATP).

Efficacy against CIN1+ irrespective of the HPV type detected in the lesion was 22.9% (96.2% CI [4.8; 37.7]) (TVC).

There was no evidence of protection from disease caused by HPV in subjects aged 25 years and above who were DNA positive and/ or with abnormal cytology at study entry.

Clinical efficacy against anal prevalent infection in women aged 18-25 years

Study HPV-009 evaluated vaccine efficacy against anal prevalent infection at the 4-year study visit. Vaccine efficacy against HPV-16/18 and against non-vaccine types HPV-31/33/45 is presented in Table 8. Cervical infection in the same women at the same visit was assessed for comparison purpose.

Table 8: Efficacy against anal and cervical prevalent infection associated with HPV-16/18 and HPV-31/33/45 in study HPV-009

		Number of women	Number of HPV- 16/18 infections	HPV 16/18 vaccine efficacy (95% CI)		Number of women	Number of HPV- 31/33/45 infection	HPV- 31/33/45 vaccine efficacy (95% CI)
Full cohort*	Anus							
	HPV group	2103	47	62.0% (47.1; 73.1)	HPV group	2103	55	49.4% (30.3; 63.6)
	Control group	2107	124		Control group	2107	109	
	Cervix							
	HPV group	2103	40	76.4% (67.0; 83.5)	HPV group	2103	76	45.2% (27.7; 58.7)
	Control group	2107	170		Control group	2107	139	
Restricted cohort**	Anus							
	HPV group	1003	8	83.6% (66.7; 92.8)	HPV group	1629	31	61.8% (42.8; 75.0)
	Control group	986	48		Control group	1684	84	
	Cervix							
	HPV group	1003	10	87.9% (77.4; 94.0)	HPV group	1629	49	51.3% (31.9; 65.5)
	Control group	986	81		Control group	1684	104	
HPV group: treatment group vaccinated with Cervarix vaccine Control group: treatment group vaccinated with modified Havrix vaccine (Hepatitis A vaccine) *Full cohort included all women with anal specimens available **Restricted cohort for efficacy against HPV16/18 infection included subjects from the full cohort with no evidence of prevalent cervical HPV 16 and HPV 18 infection or HPV 16 and HPV 18 antibodies before vaccination, who received three doses of the HPV or control vaccines. Restricted cohort for efficacy against HPV-31/33/45 infection included women from the full cohort with no evidence of prevalent cervical HPV 31, 33, or 45 infections before vaccination, and who received three doses of the HPV or control vaccine.								

Immunogenicity

Immune response to Cervarix after the primary vaccination course

No minimal antibody level associated with protection against CIN of grade 2 or 3 or against persistent infection associated with vaccine HPV types has been identified for HPV vaccines.

The antibody response to HPV-16 and HPV-18 was measured using a type-specific direct ELISA (version 2, MedImmune methodology, modified by GSK) which was shown to correlate with the pseudovirion-based neutralisation assay (PBNA).

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The immunogenicity induced by three doses of Cervarix has been evaluated in 5,303 female subjects from 10 to 55 years of age and over 800 male subjects aged 10 to 18 years.

In clinical trials, more than 99% of initially seronegative subjects had seroconverted to both HPV types 16 and 18 one month after the third dose. Vaccine-induced IgG Geometric Mean Titres (GMT) were well above titres observed in women previously infected but who cleared HPV infection (natural infection). Initially seropositive and seronegative subjects reached similar titres after vaccination.

Persistence of Immune Response to Cervarix

Study 001/007, which included women from 15 to 25 years of age at the time of vaccination, evaluated the immune response against HPV-16 and HPV-18 up to 76 months after administration of the first vaccine dose. In study 023 (a subset of study 001/007), the immune response continued to be evaluated up to 113 months. 92 subjects in the vaccine group had immunogenicity data at the [M107-M113] interval after the first vaccine dose with a median follow-up of 8.9 years. Of these subjects, 100% (95% CI: 96.1;100) remained seropositive for HPV-16 and HPV-18 in the ELISA assay.

Vaccine-induced IgG GMTs for both HPV-16 and HPV-18 peaked at month 7 and then declined to reach a plateau from month 18 up to the [M107-M113] interval with ELISA GMTs for both HPV-16 and HPV-18 at least still 10-fold higher than the ELISA GMTs observed in women who cleared a natural HPV infection.

In study 008, immunogenicity up to month 48 was similar to the response observed in study 001. A similar kinetic profile was observed with the neutralising antibodies.

In another clinical trial (study 014) performed in women aged 15 to 55 years, all subjects seroconverted to both HPV types 16 and 18 after the third dose (at month 7). The GMTs were, however, lower in women above 25 years. 470 subjects (142 aged 15-25 years, 172 aged 26-45 years and 156 aged 46-55 years) who completed study HPV-014 and received the 3 dose schedule were followed-up for up to 10 years in the extension study HPV-060. Ten years after administration of the first dose, 100% of subjects in the 15-25 years group, 99.2% in the 26-45 years group and 96.3% in the 46-55 years group were still seropositive for HPV-16, and 99.2%, 93.7% and 83.8% for HPV-18, respectively. In all age groups, GMTs remained 5- to 32-fold for HPV-16 and 3- to 14-fold for HPV-18 above those elicited in women who cleared a natural infection for both antigens.

Bridging the efficacy of Cervarix from young adult women to adolescents

In a pooled analysis (HPV-029,-30 & -48), 99.7% and 100% of females aged 9 years seroconverted to HPV types 16 and 18, respectively after the third dose (at month 7) with GMTs at least 1.4-fold and 2.4-fold higher as compared to females aged 10-14 years and 15 to 25 years, respectively.

In two clinical trials (HPV-012 & -013) performed in girls aged 10 to 14 years, all subjects seroconverted to both HPV types 16 and 18 after the third dose (at month 7) with GMTs at least 2-fold higher as compared to women aged 15 to 25 years.

In clinical trials (HPV-070 and HPV-048) performed in girls aged 9 to 14 years receiving a 2-dose schedule (0, 6 months or 0,12 months) and young women aged 15-25 years receiving Cervarix according to the standard 0, 1, 6 months schedule, all subjects seroconverted to both HPV types 16 and 18 one month after the second dose. The immune response after 2 doses in females aged 9 to 14 years was non-inferior to the response after 3 doses in women aged 15 to 25 years.

On the basis of these immunogenicity data, the efficacy of Cervarix is inferred from 9 to 14 years of age.

Duration of the immune response in women aged 26 years and older

In the Phase III study (HPV-015) in women 26 years and older all subjects seroconverted one month after the third dose. At the 84-month time point, i.e. 78 months after completion of the full vaccination course, 99.3% and 95.9% of initially seronegative women remained seropositive for anti-HPV-16 and anti-HPV-18 antibodies, respectively. All initially seropositive women remained seropositive for both anti-HPV-16 and anti-HPV-18 antibodies. Antibody titres peaked at month 7 then gradually declined up to month 18 and stabilized to reach a plateau up to month 84.

Comparison of immunogenicity of Cervarix and Gardasil

In girls aged 9 to 14 years

In a comparison trial with Gardasil (study HPV-071) in girls aged 9-14 years, superiority of the immune response elicited by Cervarix administered according to the 2-dose schedule 0, 6 months compared to that of Gardasil administered according to the 2-dose 0, 6 months and the standard 3-dose 0, 2, 6 months schedules was demonstrated for both HPV-16 and HPV-18 by ELISA (Table 9).

Table 9: Superiority assessment of anti-HPV-16 and anti-HPV-18 immune response for Cervarix (2 dose schedule 0, 6 months) over Gardasil (2 dose schedule 0, 6 months and 3-dose schedule 0, 2, 6 months) one month and six months after the last dose (Total Vaccinated Cohort)

	Antibody	N	GMT	N	GMT	GMT ratio (Cervarix / Gardasil) 95% CI (LL; UL)
Month 7		Cervarix 0,6 months		Gardasil 0,6 months		
	Anti-HPV-16	357	8256	353	4886	1.7 (1.5; 1.9)
	Anti-HPV-18	357	5268	353	1166	4.5 (4.0; 5.1)
		Cervarix 0,6 months		Gardasil 0,2,6 months		
	Anti-HPV-16	357	8256	351	4789	1.7 (1.5; 1.9)
	Anti-HPV-18	357	5268	351	1636	3.2 (2.8; 3.7)
Month 12		Cervarix 0,6 months		Gardasil 0,6 months		
	Anti-HPV-16	355	2217	347	1260	1.8 (1.5; 2.0)
	Anti-HPV-18	355	1296	347	261	5.0 (4.3; 5.7)
		Cervarix 0,6 months		Gardasil 0,2,6 months		
	Anti-HPV-16	355	2217	348	1567	1.4 (1.2; 1.6)
	Anti-HPV-18	355	1296	348	469	2.8 (2.4; 3.2)
GMT = geometric mean antibody titre by ELISA						
N = Number of subjects with post-vaccination results available						
95% CI = 95% confidence interval for the GMT ratio (Anova model - pooled variance); LL = lower limit, UL = upper limit; p-value = 0.0001						
The association between antibody levels and clinical efficacy is not fully understood						

In women aged 18 to 45 years

In a non-inferiority comparative trial with Gardasil (study HPV-010) in women aged 18-45 years, non-inferiority of the immune response elicited by Cervarix was demonstrated for both HPV-16 and HPV-18 neutralizing antibodies in all age cohorts up to three years after first vaccination (Table 10).

Table 10: Non-inferiority* assessment in terms of neutralizing antibody titres between Cervarix and Gardasil for HPV-16 and HPV-18 at Month 7 and Month 60 (ATP) in study HPV-010

		Age (years)	Cervarix		Gardasil		GMT ratio Cervarix/Gardasil 97.6% CI at Month 7 95% CI at Month 60
			N	GMT (ED ₅₀)	N	GMT (ED ₅₀)	

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Month 7	HPV-16	18-26	104	36792	103	10053	3.7 (2.6; 5.2)
		27-35	90	23908	85	4958	4.8 (3.3; 7.1)
		36-45	96	17301	83	7634	2.3 (1.5; 3.4)
	HPV-18	18-26	118	16487	131	2258	7.3 (5.1; 10.4)
		27-35	102	9502	101	1043	9.1 (6.0; 13.9)
		36-45	110	9845	91	1439	6.8 (4.6; 10.2)
Month 60	HPV-16	18-26	35	4118	40	530	7.8 (4.3; 14.0)
		27-35	43	1925	29	346	5.6 (3.0; 10.2)
		36-45	46	1784	47	765	2.3 (1.3; 4.3)
	HPV-18	18-26	39	1523	52	126	12.1 (6.6; 22.1)
		27-35	54	967	36	74	13.0 (7.6; 22.2)
		36-45	55	817	51	105	7.8 (4.5; 13.3)

ED₅₀ = Estimated Dose = serum dilution giving a 50% reduction of the signal compared to a control without serum
GMT = geometric mean antibody titre
N = Number of subjects with post-vaccination results available
 Non-inferiority was demonstrated when the lower limit of the 97.6% CI or 95% CI was greater than 0.5
 *Superiority of the immune response elicited by Cervarix was also demonstrated up to Month 60 for HPV-16 and HPV-18 neutralizing antibodies in all age cohorts. The association between antibody levels and clinical efficacy is not fully understood.

Immunogenicity in HIV infected women

Two clinical studies assessed safety and immunogenicity of Cervarix:

1. A study performed in 120 asymptomatic HIV infected females aged 18 to 25 years (61 subjects received Cervarix) in South Africa (HPV-020).
2. A comparative study of Cervarix and Gardasil performed in 257 asymptomatic HIV-infected females aged 15-25 years (129 subjects received Cervarix) in Brazil, Estonia, India and Thailand (HPV-019).

In both studies, seroconversion at Month 7 in HIV-infected subjects receiving Cervarix was 100% for both antigens. In HPV-019, seropositivity rate at Month 24 after Cervarix vaccination was 100% for HPV-16 antibodies and >96% for HPV-18 antibodies with a Geometric Mean Concentration (GMC) level more than 12 times higher than the response to natural HPV infection. In both studies, the antibody GMCs in HIV-infected subjects appeared lower than in HIV negative subjects. The clinical relevance of this observation is unknown. In HPV-019, superiority of immune responses (neutralizing antibodies) to both HPV-16 (GMT ratio = 2.74 [95% CI 1.83; 4.11]) and HPV-18 (GMT ratio = 7.44 [95% CI 4.79; 11.54]) antigens was demonstrated with Cervarix compared to Gardasil, at Month 7 in HIV-infected subjects.

The observed reactogenicity and safety profile of Cervarix in HIV-infected women was in line with the known safety profile in healthy subjects (see *Adverse Reactions*).

The vaccine did not affect the CD4+ cell count, the HIV viral load and the HIV clinical stage.

Immunogenicity in males aged 10 to 18 years

Immunogenicity in males was assessed in 2 clinical trials HPV-011 (N=173) and HPV-040 (N=556). The data showed comparable immunogenicity in males and females. In study HPV-011, all subjects seroconverted to both HPV-16 and 18 and GMT levels were non inferior to those observed in females aged 15 to 25 years in study HPV-012.

5.2 Pharmacokinetic properties

Not applicable.

5.3 Non-clinical Information

Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, acute and repeated dose toxicity, local tolerance, fertility, embryo-foetal and postnatal toxicity (up to the end of the lactation period).

Serological data suggest a transfer of anti-HPV-16 and anti-HPV-18 antibodies via the milk during the lactation period in rats. However, it is unknown whether vaccine-induced antibodies are excreted in human breast milk.

6. PHARMACEUTICAL INFORMATION

6.1 List of excipients

Sodium chloride (NaCl)
Sodium dihydrogen phosphate dihydrate ($\text{NaH}_2\text{PO}_4 \cdot 2 \text{H}_2\text{O}$)
Water for injections

For adjuvants, see section 2.

6.2 Incompatibilities

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

6.3 Shelf life

5 years.

Cervarix should be administered as soon as possible after being removed from the refrigerator.

However, stability has been demonstrated when stored outside the refrigerator for up to 3 days at temperatures between 8°C and 25°C or for up to 1 day at temperatures between 25°C and 37°C. If not used at the end of this period the vaccine should be discarded.

6.4 Storage

Store in a refrigerator (2°C – 8°C).
Do not freeze.
Store in the original package in order to protect from light.

The storage conditions are detailed on the packaging.

6.5 Nature and contents of container

Cervarix is presented as a turbid white suspension. Upon storage, a fine white deposit with a clear colourless supernatant can be observed.

0.5 ml of suspension in a pre-filled syringe (type I glass) with a plunger stopper (butyl rubber) and with a rubber tip cap, with or without needles.

Pack sizes of 1 and 10 pre-filled syringes with or without needles.

Not all presentations are available in every country.

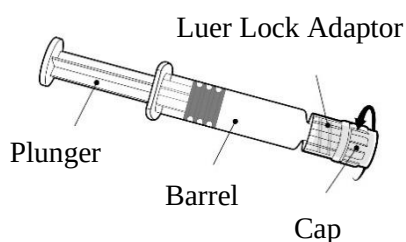
6.6 Special precautions for disposal and other handling

A fine white deposit with a clear colourless supernatant may be observed upon storage of the syringe. This does not constitute a sign of deterioration.

The content of the syringe should be inspected visually both before and after shaking for any foreign particulate matter and/or abnormal physical appearance prior to administration. In the event of either being observed, discard the vaccine.

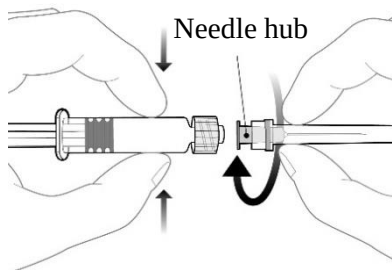
The vaccine should be well shaken before use.

Instructions for the pre-filled syringe



Hold the syringe by the barrel, not by the plunger.

Unscrew the syringe cap by twisting it anticlockwise.



To attach the needle, connect the hub to the Luer Lock Adaptor and rotate a quarter turn clockwise until you feel it lock.

Do not pull the syringe plunger out of the barrel. If it happens, do not administer the vaccine.

Disposal

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. MANUFACTURER

Manufactured by:
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Parc de la Noire Epine
Rue Fleming, 20
1300 Wavre
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